length contraction. The contraction of an object's length as a result of its motion past the person who measures the length. The contraction occurs only along the direction of motion.

light. The type of electromagnetic waves that can be seen by the human eye; see Figure P.2 on page 25.

light deflection. The deflection of the direction of propagation of light and other electromagnetic waves, as they pass near the Sun or any other gravitating body. This deflection is produced by the curvature of spacetime around the body.

LIGO. The Laser Interferometer Gravitational-Wave Observatory.

linear. The property of combining together by simple addition.

lobe. A huge radio-emitting cloud of gas outside a galaxy or quasar.

local inertial reference frame. A reference frame on which no forces except gravity act, that falls freely in response to gravity's pull, and that is small enough for tidal gravitational accelerations to be negligible inside it.

magnetic field. The field that produces magnetic forces.

magnetic field lines. Lines that point along the direction of a magnetic field (that is, along the direction that a compass needle would point if it were placed in the magnetic field). These field lines can be made evident around a bar magnet by placing a sheet of paper above the magnet and scattering bits of iron on the paper.

mass. A measure of the amount of matter in an object. (The object's inertia is proportional to its mass, and Einstein showed that mass is actually a very compact form of energy.) The word "mass" is also used to mean "an object made of mass," in contexts where the inertia of the object is important.

Maxwell's laws of electromagnetism. The set of laws of physics by which James Clerk Maxwell unified all electromagnetic phenomena. From these laws one can predict, by mathematical calculations, the behaviors of electricity, magnetism, and electromagnetic waves.

metaprinciple. A principle that all physical laws should obey. The principle of relativity is an example of a metaprinciple.

microsecond. One-millionth of a second.

microwaves. Electromagnetic radiation with wavelength a little shorter than radio waves; see Figure P.2 on page 25.

Milky Way. The galaxy in which we live.